

Inverse Hysteresis Models for Transient Simulation

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Abstract—History-dependent and history-independent models of magnetic hysteresis are proposed. Spline approximations to the magnetization curves make these models applicable to major loops of any shape. The $H(B)$ representation of reversal curves permits the development of a method that generates reversal curves guaranteed to be free from nonphysical negative slopes and excursions outside the major loop. The applicability of the models to electrical steels and a powder core is illustrated.

Index Terms—History dependent, history independent, magnetic hysteresis, phenomenological models.

I. INTRODUCTION

THE FACT that new models of magnetic hysteresis continue to be developed demonstrates that none of the existing models can be considered “universal.” The development of such a model has been impeded by the generally conflicting demands of accuracy, simplicity, and physical behavior. The latter imposes the indispensable conditions that all magnetization curves have positive slopes and remain inside the major hysteresis loop. Accuracy involves two interrelated aspects of the model. The first is its ability to describe the shape of hysteresis loops (major and minor, symmetrical and asymmetrical). The second is its ability to reproduce the magnetization history.

Often, it is sufficient to find an approximate description of the major hysteresis loop, and then to develop a procedure for constructing a trajectory, which, regardless of its starting point position, leads immediately toward the major loop tip. This trajectory is independent of the magnetization history (i.e., of the previous reversal points), so the corresponding model is called a history-independent hysteresis model (HIHM), or a model with local memory. Such models have been based on hyperbolic functions, power series, rational polynomials, or sigmoids. Examples are mentioned in [1]–[3]. The trajectories generated with these models are predetermined by the mathematics employed, and so cannot be expected to cover all shapes of hysteresis loops and all types of hysteretic behavior.

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Hysteresis modeling is severely complicated when a detailed description of the major loop is required, and the model has to reproduce the history of the magnetization process. History-dependent hysteresis models (HDHMs), also called global-memory models, have been largely developed in the framework of the Preisach approach [4] or using conceptually similar techniques in [5] and [6]. Known disadvantages of Preisach models are their sensitivity to errors in the experimental data and their relative complexity.

As an alternative to the Preisach approach, behavioral HDHMs were proposed in [7] and [8] wherein the major loop and first-order reversal curves (FORCs) were represented by splines. This representation permits these curves to be described as accurately as desired. Higher-order curves can then be obtained by copying and inserting (“transplanting”) some portion of the experimental or intuitively generated FORCs. Such transplantation-based models have been used when solving Maxwell equations [9], [10] and incorporated in transient simulators [11], [12], where magnetization conditions are not known in advance. Despite the accuracy and conceptual simplicity of the transplantation models [7], [8], their disadvantage is the relative complexity of the implementation algorithm. For this reason, when developing the phenomenological hysteresis model described here, our aim was to retain the ability of the model in [7] to reproduce a major loop of *any* shape, while simplifying the method of constructing reversal curves.

The natural basis of the proposed HDHM and HIHM is the major hysteresis loop, which can be obtained as an experimental point-to-point curve, digitized from a manufacturer’s catalog, or constructed through the use of appropriate analytical expressions. The ascending and descending branches of the major loop can then be described accurately by splines, which are library tools incorporated into the majority of computer environments. So the major hysteresis loop is not a subject for modeling. If a family of experimental FORCs is also available, the model can be adjusted to conform to these curves. In a typical situation where only the major loop is at hand, the model parameters proposed in this paper can be used.

In contrast to the majority of existing models, the magnetization curves of the models proposed here are constructed as $H(B)$ dependencies, so that the input is the flux density B , and the output is the field strength H . It will be shown that the inverse nature of such models guarantees the absence of a nonphysical negative slope (derivative dB/dH) at any point on a generated curve and keeps all hysteresis trajectories inside the major loop. The inverse form of the proposed models also simplifies their incorporation into transient simulators, where the state variables are usually magnetic fluxes or flux densities, rather than the field H .

Section II describes the principle upon which the two versions of the model are based. In Sections III and IV, this principle is used to construct the HDHM and HIHM versions. Section V demonstrates several applications of the model.

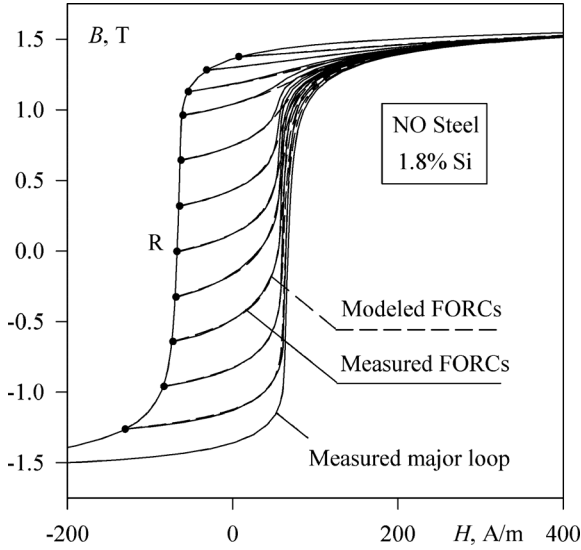


Fig. 1. Experimental major loop and FORCs of NO steel [9] (solid lines) and generated FORCs (dashed curves).

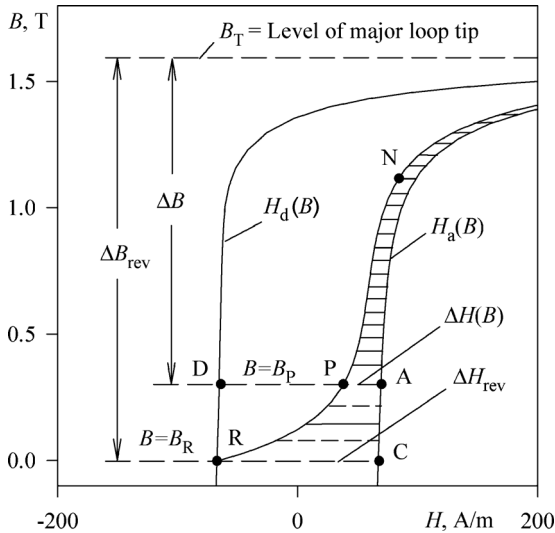


Fig. 2. Construction of an FORC R-P-N-T.

II. MODELING PRINCIPLE

The principle of the model can be explained by using as an example the construction of a FORC (i.e., a curve that starts on a branch of the major loop and ends at its tip). A family of ascending experimental FORCs of nonoriented (NO) electrical steel [9] is shown in Fig. 1 as solid lines.

Fig. 2 shows the generation of the FORC R-P-N-T that originates at point R in Fig. 1 and ends at the major loop tip T (which is outside the graph). The upper (dashed) horizontal line in Fig. 2 is drawn at the level B_T of the major loop tip T, which is the point at which the ascending $H_a(B)$ and descending $H_d(B)$ branches of the experimental major loop merge.

The first steps in constructing the sought curve R-P-N-T are to determine the width ΔH_{rev} of the major loop at the level B_R of point R

$$\Delta H_{\text{rev}} = H_a(B_R) - H_d(B_R) \quad (1)$$

and to find the distance ΔB_{rev} from point R to the loop tip T

$$\Delta B_{\text{rev}} = B_T - B_R. \quad (2)$$

At a point P on the FORC at level B_P , the field H_P can be found by calculating the distance ΔH from P to point A, which lies on the opposite (ascending) branch of the major loop. Then

$$H_P = H_a(B_P) - \Delta H. \quad (3)$$

It is seen that the gap ΔH (hatched zone in Fig. 2) shrinks from ΔH_{rev} to zero as the distance ΔB of the current point P from the major loop tip T ($\Delta B = B_T - B_P$) decreases from ΔB_{rev} to zero. For brevity, we introduce the dimensionless quantity

$$x = \Delta B / \Delta B_{\text{rev}} \quad (4)$$

which decreases from 1 to 0 as point P moves from its initial position R toward the major loop tip T.

As explained in Section III, the major loop is also referred to here as an *outer* loop. The position of initial points, such as R within this loop, will be characterized by the dimensionless ratio

$$\beta = \Delta B_{\text{rev}} / \Delta B_{\text{out}} \quad (5)$$

where ΔB_{out} is the height of the outer (major) loop.

When examining the experimental FORCs in Fig. 1, it can be seen that each of these curves first quickly approaches the ascending branch of the major loop and then its distance ΔH from this branch decreases comparatively slowly. In Fig. 2, these fast and slow behaviors take place approximately along segments R-P and P-N-T, respectively. Therefore, the distance ΔH can be written as a sum of *fast* and *slow* components represented by the first and second right-hand terms in the expression

$$\Delta H(x) = \Delta H_{\text{rev}} \cdot (1 - b)x e^{-a(1-x)} + \Delta H_{\text{out}}(B_P) \cdot b x^c \quad (6)$$

where

$$\Delta H_{\text{out}}(B_P) = H_a(B_P) - H_d(B_P) \quad (7)$$

is the width of the outer loop at the level B_P (in Fig. 2 this is the length of segment D - A). Coefficients a , b , and c are fitting parameters of the model ($a > 0$, $0 < b < 1$, $c > 0$).

The fitting can be done by taking into account the fact that the FORC can be made to recede from the major loop by increasing b . By decreasing a , the rate of decay of the fast (exponential) component in (6) is reduced, resulting in a slower decay of ΔH . This is illustrated in Fig. 3 where curves 0.5a and 5b were constructed for a twofold decrease in a and a fivefold increase of b . Parameter c controls the rate of decay of the slow component of (6).

A sequential fitting of the model to each of the experimental FORCs in Fig. 1 shows that parameters a and b in (6) depend on the position of the starting point of the FORC within the major loop, that is, on the ratio β . So the identification of the model was carried out in two stages. First, the optimum values of a and b were found for every FORC by rms fitting, and then

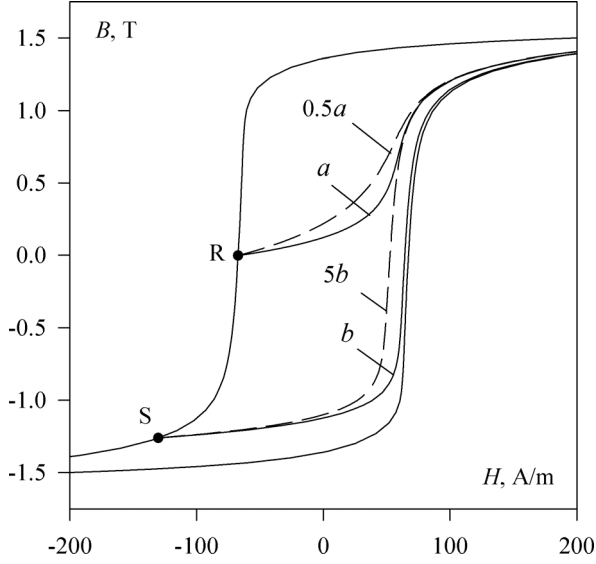


Fig. 3. Dependence of the FORC shape on parameters a and b .

an rms approximation of the derived dependencies $a(\beta)$ and $b(\beta)$ by polynomials were carried out. In the case of NO steel considered, this has led us to the following expressions:

$$a = \Delta B_{\text{rev}}(7.73 + 2.76\beta - 28.63\beta^2 + 28.36\beta^3) \quad (8)$$

$$b = 0.22(1 - \beta). \quad c = 0.125. \quad (9)$$

As curve R-P-N-T in Fig. 2, or any FORC in Fig. 1, is constructed point by point, the exponential term in (6) dies out quickly. It is important to ensure that the second (slow) term in (6) depends on ΔH_{out} , that is, on the current width of the outer (major) loop. Since $x < 1$ and x decrease along the sought curve R-P-N-T, this curve always approaches the ascending branch of the major loop and always moves away from its descending branch. This means that the slope of any constructed FORC is always positive, and that it cannot go outside the major loop. Nonphysical behavior of that kind is encountered, for example, in the models proposed in [13]–[15], wherein special corrective measures are found to be necessary.

Fig. 1 shows good agreement between the modeled and measured FORCs for the chosen NO steel. Fig. 4 shows that without making any changes, (8) and (9) are applicable to *high-silicon* NO steel [16]. The FORCs predicted with these settings (dashed curves in Fig. 4) *again* turn out to be close to the measured FORCs (solid curves).

For each specific material, the structure of (6) and the settings (8), (9) can be refined by taking into account the shape of experimental FORCs. For example, the FORCs measured for the powder core [17] (solid lines in Fig. 5) can be reproduced more accurately (dashed lines) by using $b = 0.5(1 - \beta)$ instead of $b = 0.22(1 - \beta)$ in (9).

If static FORCs are unknown, as in most practical situations, then (6), (8), and (9) can be used as given here.

III. HISTORY-DEPENDENT HYSTERESIS MODEL (HDHM)

An implicit assumption in any hysteresis model is that if its parameters are adjusted to agree with only the experimental

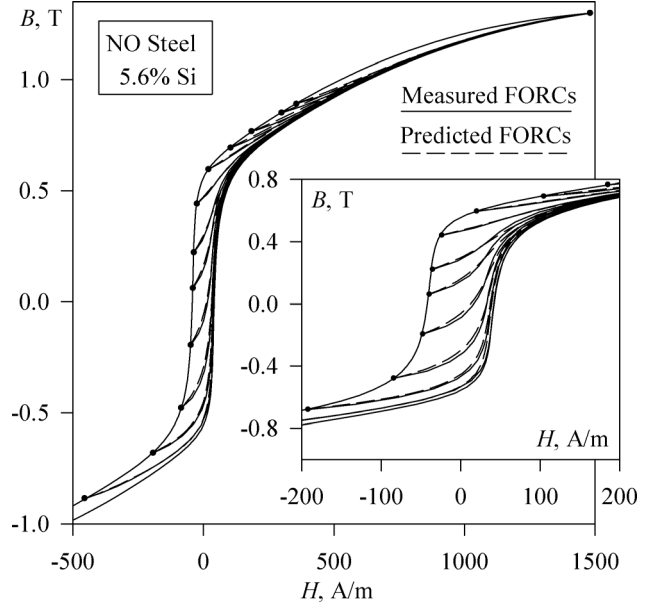


Fig. 4. Measured (solid lines) and modeled (dashed lines) FORCs of a high-silicon NO steel.

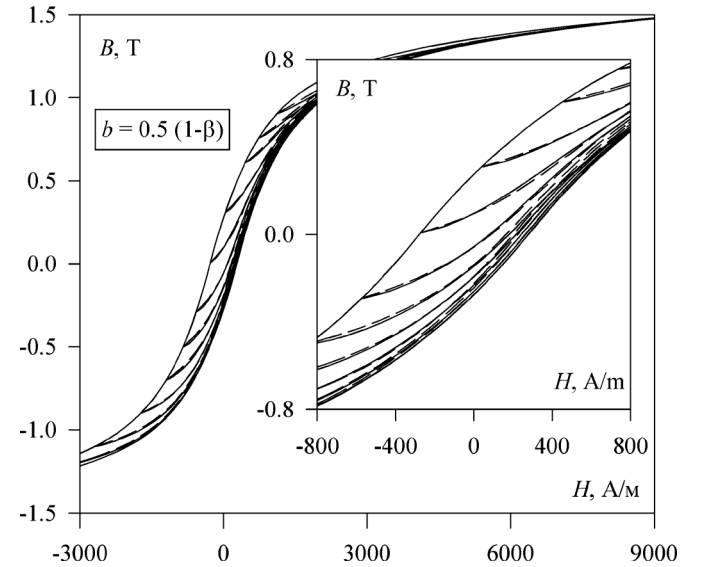


Fig. 5. Measured (solid lines) and modeled (dashed lines) FORCs of a powder core [17].

major loop and FORCs, then it will be accurate enough to predict higher-order reversal curves. Examples include the various Preisach models, the Jiles–Atherton model [1], and the positive-feedback model [18]. Following this idea, the principle described before is naturally extended to construct a *history-dependent* model. The distinguishing feature of the HDHM is that a reversal curve originating at the current turning point always leads to the previous reversal point, which is not generally the major loop tip. This behavior, also called the return-point memory (RPM) effect, was first observed by Madelung [19], and subsequently verified many times experimentally (e.g., [5]).

A basic element for constructing any reversal curve of the HDHM is its *outer loop*, which is the loop that encloses that reversal curve. In contrast to the technique described in Section II,

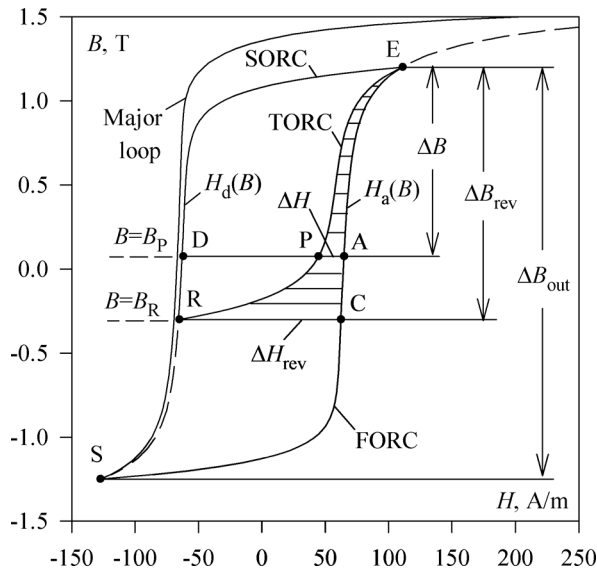


Fig. 6. Construction of a third-order reversal curve (TORC) R-P-E using the HDHM.

the *outer loop* in the HDHM coincides with the major loop only when an FORC is being constructed. In general, for a reversal curve of order N , the *outer loop* consists of the reversal curves of orders $N - 1$ and $N - 2$ (the two branches of a major loop are both zero-order curves). Due to the RPM property of the HDHM, the current outer loop is always known from the previous magnetization. It is stored in the model stack [see sequence (4) in [7]] as splines $H_a(B)$ and $H_d(B)$, representing its ascending and descending branches.

To illustrate the implementation of the RPM property, we assume in Fig. 6 that starting at reversal point S on the descending branch of the major loop, the FORC S-C-A-E and the second-order reversal curve (SORC) E-D-R-S have both been constructed using the method described here, and that we now need the third-order reversal curve (TORC) R-P-E that terminates at the previous reversal point E.

When constructing this TORC, $H_a(B)$ and $H_d(B)$ are the FORC and SORC, respectively, which form the *outer loop* for the TORC. In Fig. 6, we employ the same designations and hatched gap $\Delta H(B)$ as shown in Fig. 2. The only difference is that the *outer loop* tip E and its coordinate B_E are used in Fig. 6 instead of the major loop tip T and its coordinate B_T as in Fig. 2. Thus, Fig. 6 shows that the H -coordinate of any current point P can be calculated, as before, by means of (1)–(9). This is consistent with the idea [18] that minor loops are governed by the same laws as the major loop.

The applicability of model (6) with coefficients (8) and (9) to a material having a major loop of an *arbitrary* shape is illustrated in Fig. 7. The theoretical wasp-waisted major loop used here was originally developed for testing the transplantation hysteresis model [8]. Such a Perminvar-type loop is also typical of grain-oriented steels magnetized in the transverse direction [20].

Previously, it was difficult to obtain reversal curves for a wasp-waisted major loop using the transplantation model [8]. For this reason, [8] was confined to constructing FORCs. However, for the model proposed here, the shape of the major loop

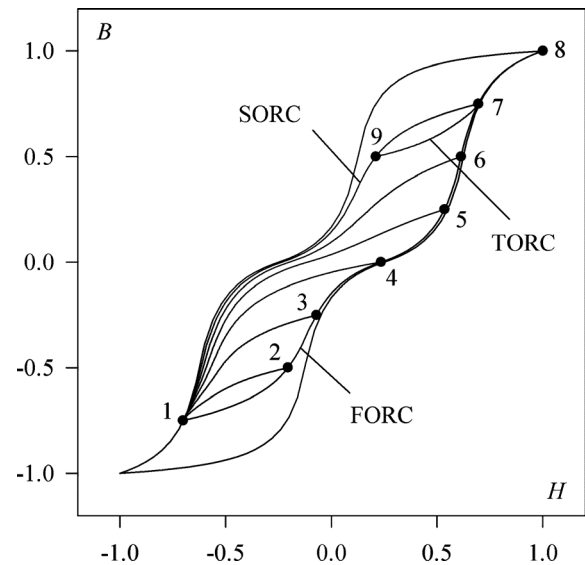


Fig. 7. First-, second-, and third-order reversal curves for a Perminvar-type major loop.

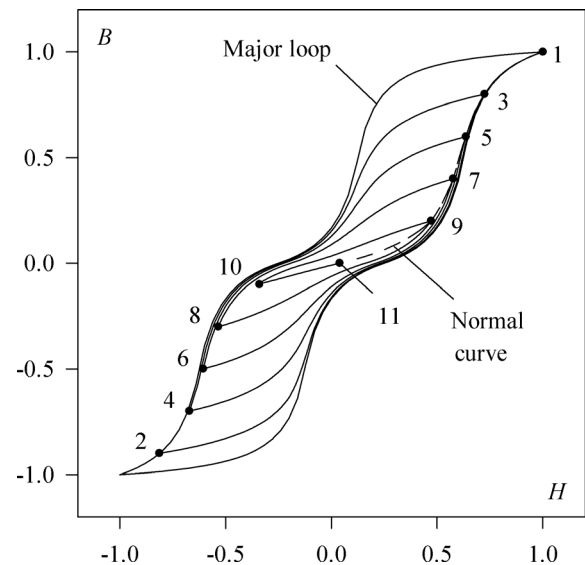


Fig. 8. Demagnetizing spiral and the normal magnetization curve.

and the order of the reversal curve to be generated are not significant considerations. For example, Fig. 8 demonstrates a demagnetization procedure starting at point 1 and ending at point 11. The B -coordinate of the penultimate point 10 can always be chosen so that point 11 is at the origin. If, having reached point 11, a magnetization in, say, the “positive” direction is initiated, then in accordance with Madelung’s rules in [7] and [19], the curve must pass through all preceding reversal points in the first quadrant (points 9, 7, 5, 3, 1), and the generated closed loops will be wiped out one after another from the model memory. If the material is magnetized in the “negative” direction, then the curve will pass through points 10-8-6-4-2.

In the FORTRAN and Matlab procedures developed, an arbitrary number of cycles of the demagnetization procedure can be employed. The spiral in Fig. 9 contains 20 cycles that are sufficient to generate a trajectory (dotted curve seen in the inset),

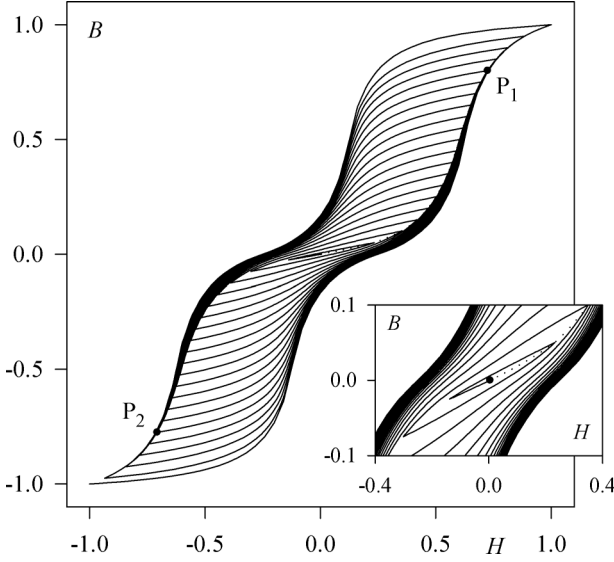


Fig. 9. Demagnetizing spiral and the associated normal curve calculated for a wasp-waisted loop, using the HDHM.

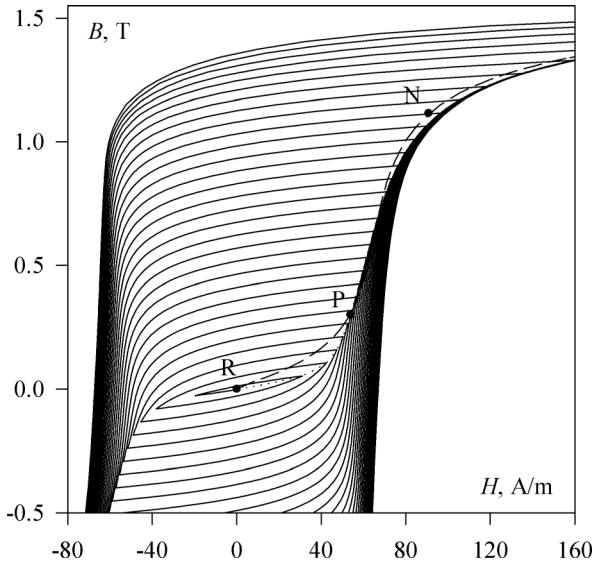


Fig. 10. HDHM demagnetization curve for NO steel [9] (solid) and the resulting normal curve (dotted) compared with the curve R-P-N constructed with the HIHM.

that starts from the demagnetized state, and stays close to the normal magnetization curve.

If the reversal is made at a point P_1 located on the normal curve in, say, the first quadrant, then the reversal curve will go to an almost symmetrical point P_2 in the third quadrant. Such behavior corresponds to the unnumbered Madelung rule in [19]. A similar demagnetizing spiral and resulting normal curve constructed for NO steel are shown in Fig. 10.

IV. HISTORY-INDEPENDENT HYSTERESIS MODEL (HIHM)

Implementation of the memory stack needed for an HDHM (see, for example, [5] and [7]) requires programming skills and may be time-consuming. However, the magnetic cores of many devices operate under a smooth excitation, such that there are no asymmetrical minor loops during startup transients or in steady-

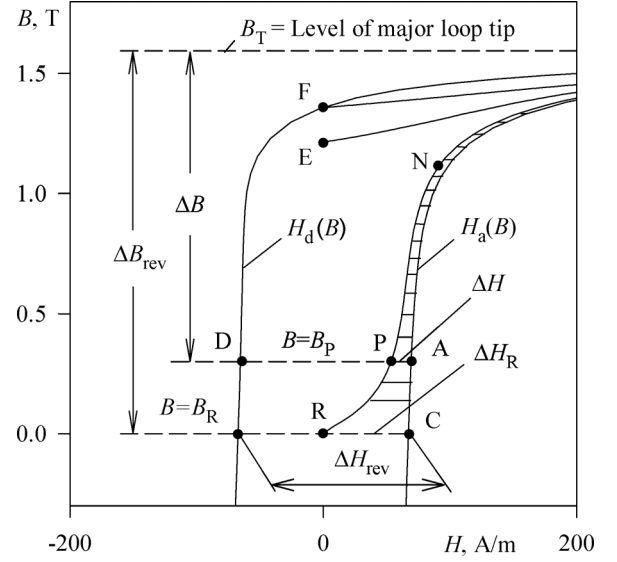


Fig. 11. Construction of a reversal curve R-P-N using the HIHM.

state regimes. In these cases, often encountered in practice, the use of the simple HIHM considered in this section is expedient.

Since the HIHM retains no information about the magnetization history (previous reversal points), the only possible endpoint for any trajectory is the tip T of the major loop, which is always the outer loop in the HIHM.

In Fig. 11, the reversal curve under construction is the curve R-P-N-T that originates from point R (H_R, B_R), arbitrarily located inside the major loop. Curves $H_a(B)$ and $H_d(B)$ are its ascending and descending branches, respectively. Although point R is here chosen to be the origin of the B - H plane, it can be the reversal point of a preceding magnetization process.

As before, the first step in constructing the sought curve R-P-N-T is to use (1) to determine the width ΔH_{rev} of the major loop at level B_R . The second step is to find the field distance ΔH_R from point R to point C on the ascending (right-hand) branch of the major loop

$$\Delta H_R = H_a(B_R) - H_R. \quad (10)$$

This value is used in (6) instead of ΔH_{rev} . The other modification in the HIHM is to introduce a scaling factor

$$\delta = \Delta H_R / \Delta H_{rev} \quad (11)$$

to decrease the “slow” component on the right-hand side of (6). As a result, (6) takes the revised form

$$\Delta H(x) = \Delta H_R \cdot (1-b)x e^{-a(1-x)} + \delta \cdot \Delta H_{out}(B_P) \cdot b x^c \quad (12)$$

where constants a , b , and c are calculated using (8) and (9), as before.

In the HIHM, the coefficient β is defined as the ratio $\Delta B_{rev} / (2B_T)$, where $2B_T$ is the height of the major loop. If the reversal point R lies on the major loop, then $\Delta H_R = \Delta H_{rev}$, $\delta = 1$, and the sought trajectory becomes an FORC.

It is shown in [3] that the use of the major loop tip T instead of the preceding reversal point can cause some distortions of

minor-loop branches and the position of the steady-state loop, but this is the price to pay for the lack of memory in the HIHM. Nevertheless, the curve R-P-N in Fig. 11 (also shown in Fig. 10) is quite close to the normal curve constructed with the HDHM. As shown in Section V, this very fact allows one to use the HIHM with almost the same accuracy as the HDHM in modeling processes in which all minor loops are centrosymmetric.

V. APPLICATIONS OF THE MODELS

It must be emphasized that static hysteresis models are the most “vulnerable” elements in finite-difference (FD) or finite-element solvers. This is because of the different magnetization trajectories in different “layers” of the sheet and the impossibility of predicting the conditions under which the hysteresis model will operate in each layer. This means that testing the model within such a solver provides an additional verification of its robustness. In particular, a negative dB/dH at any node would destroy the integration procedure immediately. To test the HDHM and HIDM proposed, they were incorporated into an FD solver [9] of the penetration equation

$$\frac{\partial B(x, t)}{\partial t} = \rho \frac{\partial^2 H(x, t)}{\partial x^2} \quad (13)$$

that determines transients in a ferromagnetic sheet of thickness d and resistivity ρ . In this solver, the partial difference equation (13) is reduced to the system of N ordinary differential equations in $B_i(t)$ and $H_i(t)$ that represent the magnetic induction and field in the i th node of the FD grid. Grid functions $B_i(t)$ and $H_i(t)$ are linked by a static hysteresis model (HDHM or HIDM) and by a differential equation reproducing the magnetic “viscosity” that represents a time delay of B_i with respect to H_i . In this way, the solver takes into account not only the static hysteresis loss but also the classical eddy-current and excess (anomalous) losses.

To speed up the approach to the steady-state regime, all nodes of the grid are initially set to a “zero state” ($B = 0, H = 0$). When using the HDHM, such a state is achieved during the pre-processing stage via the demagnetization sequence described before; the demagnetization history is then copied to each node. If the subsequent magnetization commences in the “positive” direction, then the nodal magnetic trajectories $H_i(B_i)$ will initially follow the dotted curves in Figs. 9 and 10.

Startup transients calculated with the solver for a 0.5-mm-thick NO steel sample at three different frequencies of a cosinusoidal magnetization voltage are shown in Fig. 12 by the dashed lines. The steady-state symmetrical loops already established after one period of the exciting voltage nearly coincide with the experimental dynamic loops (solid lines) measured at the same maximum induction ($B_m = 1.5$ T) [9].

We recall here that the dynamic hysteresis loop of a conducting sheet shows how the *average* induction over the sheet cross section depends on the magnetic field at its surface. This means that the accurate predictions in Fig. 12 testify to the accuracy of the static hysteresis model (HDHM) employed *independently* at every node of the grid. The difference between the nodal hysteresis loops (i.e., the loops at different depths of the sheet, from its surface to the middle) is shown in Fig. 13,

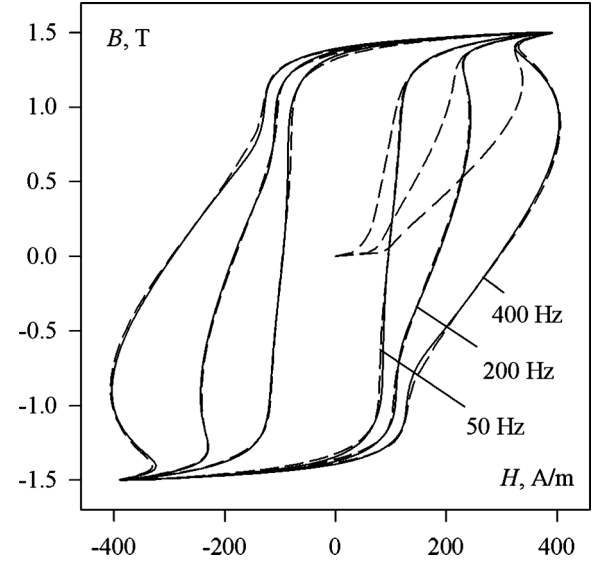


Fig. 12. Dynamic magnetization curves for NO steel at $B_m = 1.5$ T.

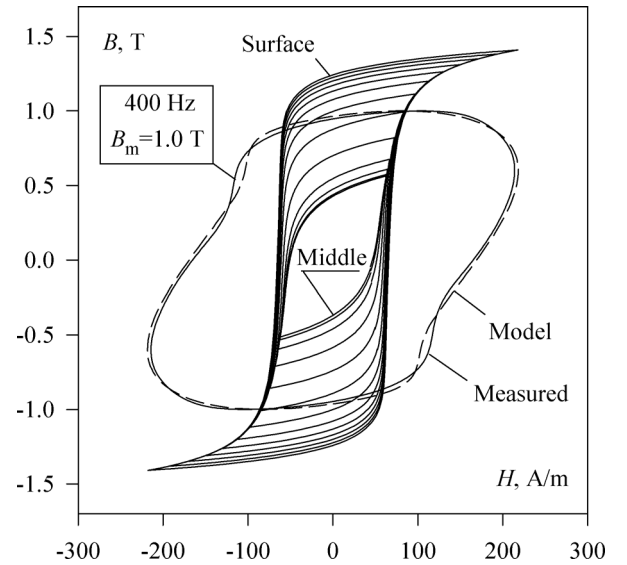


Fig. 13. Dynamic magnetization curves for NO steel at $B_m = 1$ T and corresponding nodal loops (ranging from the surface to the middle of the sheet).

which also shows that the calculated and measured dynamic loops ($B_m = 1.0$ T, $f = 400$ Hz) are again quite close.

The same good agreement is observed when the HDHM is employed for nonsinusoidal excitations. As an example, Fig. 14 demonstrates the close agreement between the predicted and measured dynamic curves for a two-level pulse-width modulated (PWM) voltage excitation. In this figure, F is the carrier frequency, f is the modulation frequency, and m is the modulation index. The calculated and measured losses are also shown.

It is remarkable that for the sinusoidal excitations used in Figs. 12 and 13, the shapes of the predicted dynamic loops and the total losses depend only slightly on whether the HDHM or the HIHM is used at each node of the FD grid. For the PWM excitation considered, the loss underestimation obtained with the HIHM is only 5%. Although this inaccuracy can be greater in

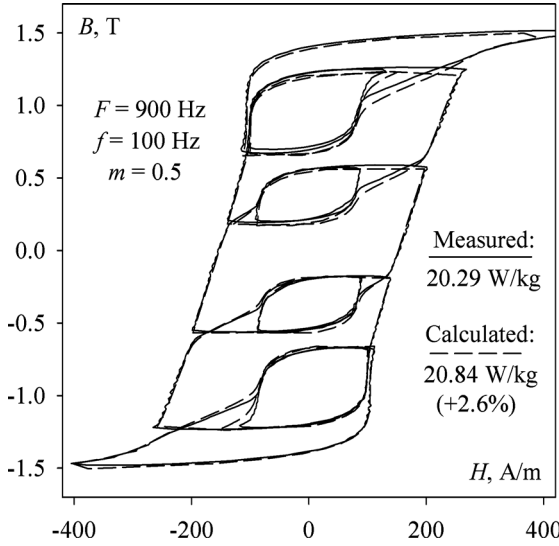


Fig. 14. Dynamic magnetization curves for a two-level PWM voltage excitation.

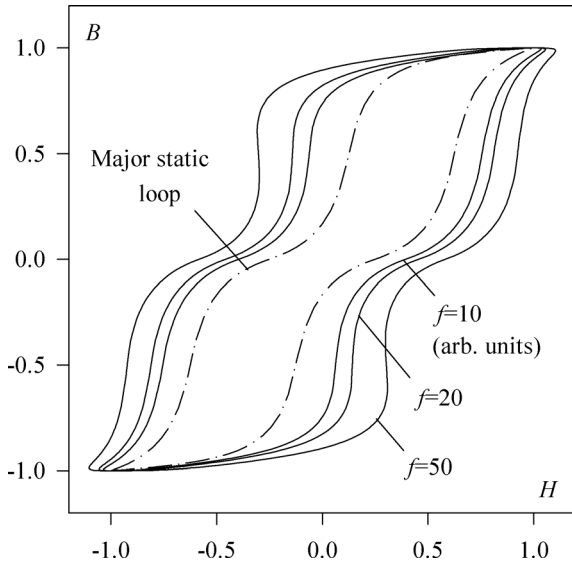


Fig. 15. Dynamic magnetization curves for a wasp-waisted material.

some cases, it is expedient to start the modeling using the simple HIHM.

It was also important to test the solver [9] for the case of wasp-waisted hysteresis loops. The stable operation of the solver, using the HDHM proposed, is illustrated by the dynamic loops in Fig. 15 obtained with the solver.

In concluding this section, we point out that (13) is primarily applicable to NO (“motor”) steels, which are characterized by magnetic domains that are much smaller than the sheet thickness.

In grain-oriented (GO) steel, which is the usual magnetic material for power and transducer transformers, the situation is different. The large domains of GO (“transformer”) steel make its magnetization mechanism quite different from that in NO steel. This reduces the frequency range over which (13) is applicable in this important type of electrical steel [10].

This justifies the direct use, in the case of GO steel, of the loss (and field) separation principle described in [21] which, in turn, leads to the thin-sheet model (TSM) [10], [16] that results in the field expression

$$H(t) = H_h(B) + \frac{d^2}{12\rho} \frac{dB}{dt} + g(B)\delta \left| \frac{dB}{dt} \right|^{0.5}. \quad (14)$$

Here, $H(t)$ is the magnetic field at the sheet surface, $H_h(B)$ is the field calculated by means of a static hysteresis model (HDHM or HIHM), $\delta = \pm 1$, and the function $g(B)$ serves to make the calculated dynamic loop agree with the measured loop [12].

The common feature of the TSM (14) and the FD solver [9] is the use of the inverse hysteresis dependencies $H(B)$, which are reproduced by the models proposed here.

VI. CONCLUDING REMARKS

We propose two simple but flexible hysteresis models, one with local and the other with nonlocal memory. The phenomenological nature of the models should not be considered a disadvantage, particularly in view of the fact that certain models claimed to be physical are, in fact, phenomenological ones [22].

Unlike the Preisach models, the models proposed here should not be considered mathematical operators that, given an input function $B(t)$ generate an output function $H(t)$. Instead, they are computational algorithms in which a number of steps leads from $B(t)$ to $H(t)$. In this sense, the models described in the paper are “equation free.” We do not maintain that the model settings (6), (8), and (9) are the only possibility. When experimental FORCs are also available, the parameters of the model can be fitted to these FORCs. In the usual situation where the user has only the major loop, the model settings provided by this paper can be employed.

The inverse nature of the proposed models does not preclude their use for predicting $B(H)$ curves. Since a modeled magnetization curve is represented as a set of points in the B – H plane, it can be described by $H(B)$ and $B(H)$ splines simultaneously.

A user-friendly demo version of the models is available at <https://sites.google.com/site/inversehysteresismodel> where major loops of different materials are provided. Users can also prepare their own major-loop files to test the proposed algorithms. In the demo, the $H(B)$ curves are extrapolated beyond the major loop tip as straight lines with a slope equal to the half sum of the slopes of the major loop branches at the tip (if these slopes are different). A more accurate extrapolation of the major loop to the saturation (nonhysteretic) region can be carried out in accordance with [23].

In addition to the FORTRAN and Matlab procedures developed, the hysteresis models described here have been recently incorporated in the Electromagnetic Transient Program–Alternative Transients Program (EMTP-ATP). Some details of the implementation will be published elsewhere.

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